

Dog Skin Disease Recognition Using Image Segmentation and GPU Enhanced Convolutional Neural Network

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Abstract—Some, if not all, veterinary clinics do not have a record of skin diseases of dogs when they diagnosed them; this is due to lack of manpower over the number of different kinds of animal patient that they cater per day. This also causes some delays in diagnosing other non-visually diseases that other patients might have. Having a system that can be used in pre-examination for visually available infections such as dog's skin disease and automatically records this diagnosis, can give an advantage to veterinary clinics. This helps the clinics prepare for the kind of patients that they will tend in certain time of the year. Developing a system that can be used in identifying common dog skin diseases for the pre-examination purpose and creating a dashboard that generates a numerical result can be used as an advantage for the veterinary clinics. These can be achieved by using image processing techniques for the prediction model and Convolutional Neural Network (CNN). However, using a common CNN approach, where the main core uses the Central Processing Unit (CPU), tends to train the model longer. To overcome this problem, the use Graphics Processing Unit (GPU) is implemented to enhance the speed of training the model for the system. Having this kind of system really helps the veterinary clinic for their daily work, but this can still be improved by using other approaches to the trained model without ignoring the efficiency and accuracy of the algorithm that is being used.

Keywords—CNN, CPU, GPU, Image Processing, Skin Disease

I. INTRODUCTION

Dogs are one of the most common animals that we can see everywhere, like in households, police, military, or even in dog shows [1]. They provide protection to our households, and sometimes is the best human companion [2], especially for persons with disabilities. They are called "man's best friend" [3] for that reason. However, like any other animals, dogs are not immune to a skin disease. Their skin disease is caused either by allergies, irritants, immune system-related problem, or other factor, such as mites. If these diseases are not identified and

treated correctly, it can affect how the dogs behave and its health negatively. It may even spread to other dogs or worst, human beings. As technology evolves, different kinds of applications and systems have already been developed, especially for medical approach [4-7]. However, these applications are mostly available for human health purposes only. Animals are still being sent to a veterinary doctor for checkup where, not all the time, there is an available veterinary clinic or veterinarian to cater them.

This study aims to create an application which will prioritize animals, such as dogs, for quick and reliable diagnosis. It will benefit the volunteers in animal shelter organizations who caters stray, abused, abandoned, and senior dogs that are most susceptible in skin diseases. This study also aims to pre-examine the situation of the dog particularly to newly rescued dogs, in which there might not be an available veterinarian volunteer to check the dog's condition. It can also be beneficial to the veterinarian in terms of additional information of the dog condition after the volunteer uses the application for an on-site check.

A. Purpose and Description

Veterinary clinics caters to multiple kinds of animals per day. They also cater to certain number of scheduled check-ups or emergency cases that requires special treatment and specialized veterinarian. Unfortunately, not all veterinary clinics have enough manpower to handle these daily number of patients. This tends to delay the patients that causes the rescheduling of check-ups. Due to delay of response, the record keeping, or logging of dog's skin diseases are delayed as well or not recorded at all. In most cases, veterinary clinics don't have a record of dog's skin disease they diagnose, which these records can be used in studies and prediction of most common skin disease for the certain months. Therefore, this research help creating a tallied record of detected skin diseases of dogs using

a simple system that both doctors and veterinary clinic personnel can use for pre-examination and recording purposes.

B. Project Context

Despite the availability of veterinary clinics in different places, not all dogs are getting checked-up or diagnosed on time. The length queues in the clinic are long because of the variation of pets with different needs, availability of the veterinarian and financial issues of the dog owner. With these not all dog who queue gets administered and the severity of the infections are not handles immediately.

C. General Objectives

The researchers aim to develop a web and mobile application for veterinary clinics that help them identify the 12 skin diseases of dogs using image processing and classification techniques.

Specifically, the research purpose was to design and develop an interface that will identify these skin diseases using TensorFlow's implementation of Convolutional Neural Network with the incorporation of Graphics Processing Unit as support. Another aim was to create a dashboard that generates numerical occurrences of a certain skin disease, to validate and test the application for the accuracy of the trained model and to implement a system that will help a veterinary clinic to cater dogs in case of unavailability of veterinarian for simple, fast, and reliable dog skin disease diagnosis.

D. Scope and Delimitation

The study will focus on developing a system that will help the veterinary clinics in pre-examination dog patients particularly focusing on skin diseases of dogs. The system has a mobile and website application that rapidly processes the image of the infection found in the skin of the dog. This application is not substitute for any professional diagnosis done by the veterinarians. It would provide an additional help in pre-examination of potential diagnosis. The diagnosis that was detected will also serve as an additional information to the veterinarian's diagnosis.

II. REVIEW OF RELATED LITERATURE

A. Local Literature

Dog skin diseases, if not treated early, can be very dangerous for dogs [8]. Sometimes most of the people do not give much attention to their dog's skin problems and treat them using their own method not knowingly how dangerous their pet's skin diseases are. Treating dog skin diseases at its initial stage is important to prevent it from spreading.

The mobile application will help the users to identify twelve dog skin diseases yeast infection, folliculitis, impetigo, seborrhea, ringworm, alopecia, mange, fleas, color or texture changes, acral lick/granuloma, skin tumor, hot spots and anal sac diseases each subtype which will be discussed.

Yeast is a fungus that lives on the skin and inside the gut of your dog without causing any harm, most of the time. Folliculitis is a term that refers to the inflammation of one or more hair follicles. Impetigo is a benign, non-contagious, non-zoonotic superficial bacterial infection of skin that does not involve hair follicles. Seborrheic Dermatitis (SD) of the scalp is characterized by scaly, reddish lesions. Ringworm, despite its

name, it is not an actual worm but a fungus that is highly contagious to other animals and humans as well. Alopecia can affect dogs in all breeds and genders at any stage of their life. It is a common condition and can display either partial or complete hair loss, but it can also affect the dog skin, endocrine, lymphatic, and immune system of the dog. Allergic Dermatitis or dogs have allergies reacting to the food they eat, and environmental irritants like pollen or insect bites. Mange is a skin disorder caused by tiny parasites called mites. Changes in the skin color of dogs is another skin disease that is because of common metabolic or hormone problems. Skin tumor or a hard lump on the skin of the dog. Hot spots, also called acute moist dermatitis, are small areas that appear red, irritated, and inflamed. And lastly, anal sac disease that affects the anal sac of the dog that would lead to foul smell or itchiness of the area [9].

B. Foreign Literature

CNN dominates the results on varying image classification tasks. In which in the study of Bakato and Radosav (2018), they reviewed more than 300 studies and narrowed it down to 46. Wherein these 46 studies indicates that CNN is widely used when it comes to deep learning and medical image analysis [10]. It is also backed up by using the CNN together with GPU in the test conducted by A. Krizhevsky et al. [11] where they used the GPU as the main core of the CNN to accelerate the learning speed of the ImageNet.

C. Local Studies

As automated medical diagnosis progress, some of this are already being incorporated or studied in human cases. For instance, the SkinCheckr study of the students of Far Eastern University – Institute of Technology, where they conducted and deploy a system where it can detect a human skin disease using Central Processing Unit (CPU) based image processing. However, this was only being used for human cases only and there is no available similar system for veterinary purpose [12].

D. Foreign Studies

Presented that the usage of GPU in enhancing the performance of the classification of the neural network using the vertex and pixel shader of the GPU, wherein the limitation that he experiences with the outdated architecture of DirectX APL has been overcome by using the new unified shader of the Compute Unified Device Architecture implementation. It also been noted that the ideal parameters of neural network (network structure, initial value range, learning method, learning rate, etc.) can only be determined by the testing and evaluation were shortening the training time often leads to a better result [13].

III. METHODOLOGY

A. Operational Feasibility

The system is implemented through mobile app and web application and was named *Petment Scanner*. The system's goal is to help the management of skin diseases in veterinary clinics. It helps fasten the consolidation of the different skin diseases of dogs and at the same time gives a reliable pre-examination of dogs before the actual diagnosis test. The system will also provide a record of the specific skin diseases that prone to appear within a certain season of the year.

B. Technical Feasibility

All users can use both android devices and computers to use the system but mobile device is the most direct approach for the capturing and uploading of the image. The minimum requirement for the mobile device is an android operating system with version 5.0 or higher while for computer is at least with 4GB RAM and a processor of Pentium 4 or higher. In order for the administrator to update the trained model, they need to have at least 8GB RAM, Intel i3 processor and Nvidia GTX 1050ti for the GPU in order to retrain the model of the system.

C. Economic Feasibility

The person who will be using the system will find it very easy to use and understand in this way, the management can still focus in their usually daily checkup and diagnosis while the front desk of the clinic can use the system as the initial checkup.

D. Project Design

There were three algorithms that were implemented in this research, which is inspired by recent studies in image processing techniques [14-16] and machine learning [17, 18]. First is the image segmentation algorithm is used. This simply converts the image into something meaningful and easier analysis. Here the researchers used the thresholding method where in it converts the gray-scale image into a binary image. As shown in Figure 1, the segmentation process produces sets of segments that covers the entire image, shows the identified pixels and computed property of the color, intensity, and texture of the image.

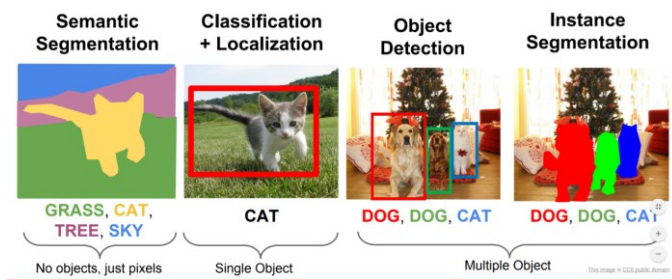


Fig. 1. Different kind of Image Segmentation

Secondly, Otsu Algorithm thresholding that can perform a simple image thresholding. This algorithm returns a single intensity threshold that separates pixels into two classes, namely the foreground and the background. This algorithm was used to separate the objects in the foreground with the object found in the background of the image as shown Figure 2.

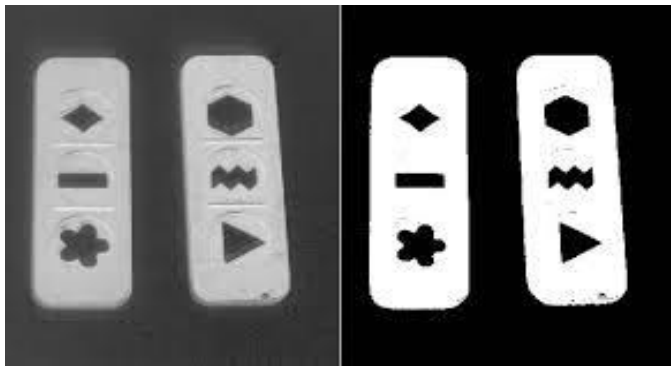


Fig. 2. Otsu's Thresholding foreground and background separation

And lastly, the Convolutional Neural Network that was used to classify the targeted skin disease using its classic architecture of layer. This process separates the non-linearity in the image and removes negative values from the activation map to obtain the targeted data in the image as shown in Figure 3. Based on the research findings conducted by Buber and Dirir (2018) where the result of their test shows that the use of GPU core consumes less time of training specially if there are a great size of dataset batch [19], which pushes researchers to utilize the use of GPU core instead of CPU to maximize the time CNN usually uses CPU for its deep learning process that requires a big system load to the CPU to execute. This takes a lot of time to train the model, however, using Nvidia GTX 1050ti GPU as the core of CNN using the Compute Unified Device Architecture API from Nvidia for parallel performance, makes the training time of the model much faster thus shortens the time of training the model.

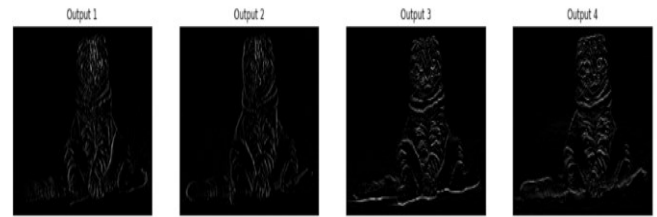


Fig. 3. CNN output per process of Rectified Linear Unit (ReLU)

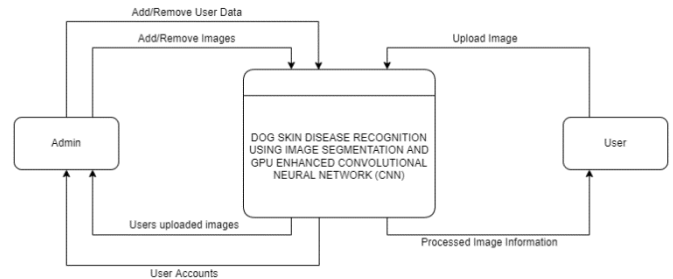


Fig. 4. Context Diagram

To implement this in the application, Figure 5 shows the function of two direct groups with the system. The admin can add or remove both user account and all uploaded image by the user in using the system, they also are the responsible for updating the trained model for a longer run and more precise prediction while using the system. This administrators are the management of the veterinary clinics who will utilize the system in their facility. The users such as the front desk of the clinic and dog homeowners can create an account, capture, and upload an image and view any data of their own uploaded images to the system. The uploaded image can be viewed by administrators. The system can provide prediction of what dog skin diseases were detected based on the count or the use of the application.

E. System Architecture

The users will have an option to run the application either on mobile or on computer devices, while the admin will need to access this using a computer device. Through the devices, the application will run either to send or request over the internet to the database to upload, view or manage data that the users sent. The database fetches the data from the application and save it in

both web a mobile system. Information of the processed images will be updated to the users view and the total counted record of predicted dog skin diseases will be available to all users as well.

F. Data Gathering

The researchers used a quantitative approach through surveys. This was used to collect the required data that helps to enhance the system. This approach is used by the researchers to gain insights of the performance of the system. The comments and suggestions of the users were used to help enhance the system. Any additional information received also helped in the creation of an effectiveness system.

G. Sampling Technique

The researchers utilized a combination of convenience and purposive sampling, due to the availability of the veterinarian and veterinary clinic personnel. The respondents were required to answer a survey after they have used the prototype. There were 5 developers or instructors, 5 registered veterinarians, 13 dog owners, and 7 experienced clinic personnel who tried the prototype and answered the survey.

H. Statistical Treatment

The statistical treatment used by the proponents to gather data is the weighted mean. The weighted mean formula was used to interpret the data from the surveys collected. The purpose of calculating data is to acquire accurate results. To find the weighted mean average, the proponents used the formula below:

$$\text{Weighted mean} = \frac{\sum wx}{\sum w} \quad (1)$$

Where:

w = weighted average

x = value of the items

IV. RESULT AND DISCUSSION

The software's reliability was evaluated through a series of test case with at least ten trials per run, after training the model. The first test evaluates the application to detect healthy skin only. The second test is for the ability to detect the scoped skin diseases. Third test is a combination for first and second, finally the fourth and last test was to evaluate the application's ability to detect non-skin objects. Based on the tests, a 70% reliability resulted on images of healthy skin, 90% reliability resulted on images of non-healthy skin, 70% reliability resulted on images of healthy and non-healthy skin, and 60% reliability on images of non-skin images (Figure 5). The detection of whether the skin is healthy or not is another feature of the system.

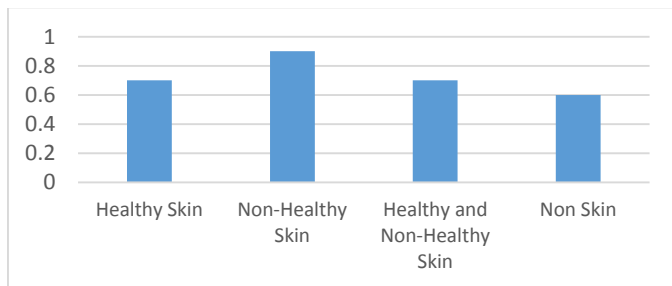


Fig. 5. Reliability of test results

The system also has the means to detect what skin disease a certain canine has. A reliability test was conducted to the application, and it returns a result of specific skin disease that it is detected as shown in figure 6.0. In the figure shows the different types of skin diseases detected such as Acral Lick Dermatitis, Alopecia, Anal Sac, Ringworm, Folliculitis, Hot Spot, Impetigo, Yeast Infection, Tumor, Seborrhea, Mange and Allergic dermatitis. The figure report shows that there are a lot of Tumor diseases found in dogs. This data is based on accredited clinic that the researchers have partnered with.

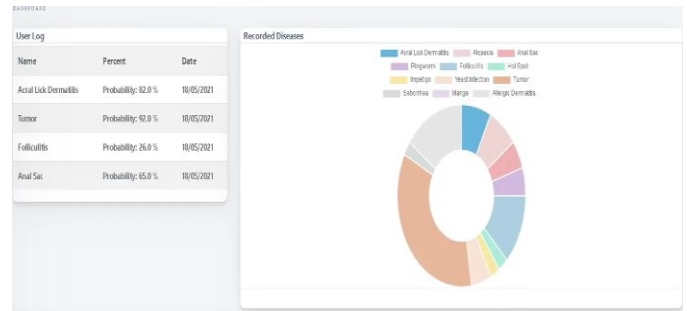


Fig. 6. Application tests results

The system was then given to the users to test. Based on the results found in Table 1, the result of the survey shows that the system has resulted to a weighted mean of 4.6 (90%) for functionality which means that the application's ability to serve its purpose was above average, 4.5 (90%) for usability where it is easy to learn and use, and 4.6 (92%) for efficiency, meaning that the application is stable enough to response efficiently. The total average of application named *PetMent Scanner* weighted score was 4.5 (90%), functioning and delivering its main objective of detecting and diagnosing image of diseased dog skin in above average.

TABLE I. SUMMARY OF FINDINGS

Category	Weight Mean	Interpretation
Functionality	4.6	Outstanding
Efficiency	4.6	Outstanding
Usability	4.5	Above Average
Total Average	4.5	Above Average

The researchers utilize the GPU core instead of CPU to maximize the time of training the system as shown in figure 7 wherein it takes 15 seconds for 1 whole 9 steps epoch run per 1 dataset training. This presents a more robust and faster way of performing all the tasks necessary to achieve the goal of the system. With each iteration, the loading speed can build up and it is only necessary for developers to find a way on how to secure a faster way and develop a more user-friendly application not only for this application but also in other health systems [20-24].

Name	Smoothed Value	Step	Time	Relative
Dogs-Skin-AI-CNN\train	0.2443	0.1139	9	Wed Mar 3, 02:01:16 15s

Fig. 7. Total time of Epoch run per dataset

V. CONCLUSION

Based on the findings of the present study, the following conclusions were drawn. The researchers have developed a system for detection and diagnosis of dog skin diseases using image processing techniques and achieved in creating a model with the application of Convolutional Neural Network (CNN) enhanced by Graphics Processing Unit (GPU) for an enhanced maximum speed of training. The speed of training with the use of Nvidia GTX 1050ti GPU as the core of CNN is at the optimized rate than the usual CPU core CNN, however it can still be enhanced by using the latest version of CUDA supported GPU that have a higher score level. The application *PetMent Scanner* has achieved its goal of utilizing the trained model of neural network in classifying different dog skin diseases after undergone an image processing. If applicable, differential diagnoses are shown to show the user of other possible diagnoses. To further help the veterinary clinic in keeping a record, a dashboard featuring the tally of detected dog skin diseases has been added. The diagnosed image can be saved in the user's local memory in mobile phone and through online database server

VI. RECOMMENDATION

Based on the conclusion, the following recommendations are offered. For future researchers, the researchers recommend that the study can be improved by including different breed of the dogs that can be added to be classified in neural network, as well as more images in the training and test model for more accuracy. Using a higher version of GPU that supports a higher score rate of Compute Unified Device Architecture (CUDA) may be used to achieve a different result in speed of the training process. Different type of classification model such as Artificial Neural Network (ANN), may be used together with CUDA for different result and accuracy. Alternative or additional image processing techniques such as OpenCV for the real time detection are recommended to be implemented, but the efficiency of the algorithms should not be neglected.

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